

# Jeremy Lopez

JRLopez8009@gmail.com | Mobile: (786) 830-9988 | <https://www.linkedin.com/in/jeremylopez1/>

## EDUCATION

### University of Central Florida

Bachelor of Science in Mechanical Engineering — GPA: 3.4

Expected Graduation: 2028

## EXPERIENCE

### Project Lead — ASME Student Design Competition, UCF

Sept 2026 - Present

- Lead a multidisciplinary student team through the full design, build, and compete cycle of a single ASME-compliant robot for the CY2027 “PODIUM” season of the ASME Engineering Games, competing across five scored disciplines, including The Sprint, The Lift, The Cycle, The Marksman, and The Ascent.
- Direct system-level design trade-offs and subsystem integration (drivetrain, manipulators, climbing mechanism) to balance competing performance demands, including speed/agility, power, throughput, accuracy, and climbing, within one competition-legal robot platform.
- Own the team schedule, budget, and inspection readiness, and coordinate the engineering design report and technical documentation submitted for competition judging, while serving as primary liaison to the faculty mentor and event officials.

### Research Assistant — Interfacial Heat Transfer Laboratory, UCF

June 2026 - Present

- Design and iterate microchannel geometries for a coolant-cooled aluminum nitride heat exchanger, engineering and testing multiple channel configurations, varying width, spacing, and pattern, to determine which geometry maximizes heat dissipation for high-heat-flux ( $\geq 100 \text{ W/cm}^2$ ) power electronics cooling, under the guidance of faculty advisor Dr. Shawn Putnam.
- Research and synthesize peer-reviewed literature and experimental data on liquid-cooled microchannel heat exchangers, covering channel geometry, coolant flow distribution, and substrate thermal properties, to ground design decisions in validated results.
- Develop SolidWorks CAD models of candidate microchannel heat exchanger geometries and prepare technical presentations and documentation communicating design rationale, governing equations, and comparative thermal performance data to faculty.

### Drive Team Lead — ASME Student Design Competition

Sept 2025 - April 2026

- Developed a SolidWorks CAD model for a competition-compliant waste-sorting robot integrating color sensors.
- Integrated electronics, including motors, controllers, and power systems, with mechanical design, validating performance through cross-disciplinary team testing and iterative design reviews.
- Gained hands-on experience in cross-disciplinary teamwork, project management, and competition.

### Resident Assistant — UCF Housing & Residence Life

May 2026 - Present

- Support a community of students by fostering a safe, inclusive, and academically focused living environment.
- Serve as a first point of contact, providing guidance, conflict resolution, and referrals to campus resources.
- Partner with housing staff to plan and execute community programs and enforce residential policies.

## PROJECTS

### Arduino-Based Auto-Irrigation System

Jan 2026 - Feb 2026

- Built a low-power automated plant watering system using an Arduino Nano, soil moisture sensor, and relay-controlled pump.
- Engineered and tested a single-supply 5V power architecture with surge suppression and power stabilization, validating reliable operation under continuous motor load.
- Modeled and 3D-printed a custom enclosure integrating a reservoir, electronics housing, and plant pot into a portable, field-tested unit.

### Intro to Engineering - Great Navel Orange Race

Jan 2026 - April 2026

- Designed and fabricated an autonomous pond racing boat using laser-cut water-resistant wood, incorporating custom CAD-designed DXF ribs and hull structures for durability and performance.
- Integrated an Arduino Nano-based control system with a brushless motor and ESC, testing propulsion, maneuverability, and control response under competition conditions.
- 3D-printed a custom rudder assembly and waterproofed the vessel, validating environmental durability using CAD/CAM manufacturing and rapid prototyping techniques throughout the design cycle.

### Intro to Engineering - Catapult Project

Aug 2025 - Dec 2025

- Collaborated with a 4-member team to design and test a latch-triggered elastic catapult robot for a school-sanctioned competition.
- Implemented servo and Arduino electromechanical integration, iteratively testing to optimize launch accuracy and consistency, and earned 1st place out of all competing teams.

## ACTIVITIES & LEADERSHIP

### SolidWorks Student Leadership Program

Sept 2025 - Present

- Complete SolidWorks workshops and assignments while utilizing SSLP resources and practice tests to pursue industry-recognized CSWA and CSWP certifications; engage in team-based design projects and technical workflows.

### UCF Brazilian Jiu-Jitsu Club

June 2025 - Present

- Active member of the university Jiu-Jitsu club, training and competing consistently while balancing academic and extracurricular commitments.
- Collaborate with teammates during training sessions to develop strategies and refine techniques, emphasizing teamwork, discipline, and problem-solving.

## SKILLS

**Software:** Excel, PowerPoint, Word, SolidWorks, Tableau

**Interpersonal:** Communication, Leadership, Problem-Solving, Project Management, Adaptability

## AWARDS & HONORS

- 2024 and 2025 AP Scholar with Distinction
- Founding Secretary, Marjory Stoneman Douglas High School Physics Club